

Report No.: 002379R		Report Date: 09/08/2024
Customer Name:	SAVITA OIL TECHNOLOGIES LIMITED	
Contact Person:	Mr. Girish Morde	
Address:	A 2/1, 2/2, MIDC INDUSTRIAL AREA, MAHAD, RAIGAD,	
	MAHARASHTRA-402309	
Sample submitted as:	Sample x 1	
Description on label/Sample Bottle:	TRANSOL SYNTH – 100	

Sample Number	18071214	Date of Sample Recieved:	10/06/2024
Date of Job started:	10/06/2024	Date of Job Completed:	08/08/2024

Project Description

A study was performed to assess the adverse effects of TRANSOL SYNTH.100 on daphnids using the species *Daphnia magna* as per OECD 202 (*Daphnia* sp., Acute Immobilisation test). *Daphnia magna* Straus belongs to the family of Crustacea, which is a typical part of zoo-plankton. Young daphnids, aged less than 24 hours at the start of the test, are exposed to the test substance at a range of concentrations for a period of 24-48 hours. Immobilisation is recorded at 24 hours and 48 hours and compared with control values.

Sample Description:

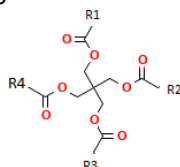
Product name : TRANSOL SYNTH.100

Type of product : Synthetic Ester based Dielectric fluid

CAS number : 68424-31-7

Sample Provided By: SAVITA OIL TECHNOLOGIES LIMITED

Structural Formulae: Pentaerythritol esters



Where,
R1, R2, R3 and R4 are alkyl groups may be same or different chain length.

Appearance and odor : Pale Yellow liquid

Physical state : Liquid

Vapor pressure (mm Hg) : <0.01 mm Hg at 200°C

Solubility in water: < 1mg/L

Stability in light: Stable

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SUMMARY

The objective of the study was to determine the 48-hour acute toxicity of against the mobility of the freshwater planktonic *Crustacean*, *Daphnia magna*.

The study was conducted in accordance with the requirements of OECD Chemicals Testing Guideline No. 202 *Daphnia* sp. Acute Immobilisation Test (adopted April 13, 2004).

The test substance was a clear liquid. Solubility data provided by the customer indicated that the test substance was insoluble in water hence polysorbate 80 emulsifier was used as per guidelines and a blank was run along with the test substance.

The test was conducted using a static test design for over a 48-hour test period

Based on the results obtained the definitive test was conducted at the following concentration - 5mg/L, 10mg/L, 20mg/L, 50mg/L, 100 mg/L.

At the start of the exposure, five juvenile *Daphnia magna* were added to each test vessel. The *Daphnia magna* in each test vessel were observed at 24 and 48 hours. The *Daphnia magna* was not fed during the test. The test concentrations were visually colourless solutions at the start of the test. No precipitation of test substance was observed during the test.

The validity criteria for control immobility ($\leq 10\%$) and dissolved oxygen (≥ 3 mg/L) were both satisfied. The test is therefore considered valid.

A range-finding test was performed until 1000 mg/L to determine the range of concentrations for the definitive test, as no Immobilisation was observed limit test was performed using 5 mg/L to 100 mg/L. No effect on *Daphnia magna* was observed.

The following results could be determined for the test item TRANSOL SYNTH.100 (species: *Daphnia magna*)

24 h-NOEC = 100 mg/L

48 h-NOEC = 50 mg/L

24 h-EC50 = >100 mg/L

48 h-EC50 = >100 mg/L

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Procedure-

Preparation of Test Sample and Control concentration:

The test item was insoluble in water hence to maximize the amount of test item in solution, the test solutions were prepared in dilution water by adding 5.0mg/L, 10.0mg/L, 20 mg/L, 50mg/L, 100mg/L weight of the sample and 10 mg of Polysorbate 80 used as emulsifier for dissolution. The Blank control was prepared by adding the same amount of Polysorbate 80 as used in test solution. The reference control Potassium dichromate K₂Cr₂O₇ (Lot No - 222403A, Sigma Aldrich- Merck, CAS No-7778-50-9) was used. A stock solution containing 250 mg K₂Cr₂O₇ /L in deionised water was diluted with dilution water to give a series of concentrations ranging from 0.5 to 2.5 mg/L.

Test Organism

Specification

Species	<i>Daphnia magna</i>
Variety	Straus
Origin	INDIA

Culture Conditions:

Test Vessels	Glass beakers of 100ml capacity
Medium	M4-Medium
pH	7.6-8.0
Food	<i>Scenedesmus abundans</i> (Green algae)
Medium renewal	Twice a week
Photo periodically	16 hrs of light 8 hrs of darkness
Temperature	20 ± 2°C

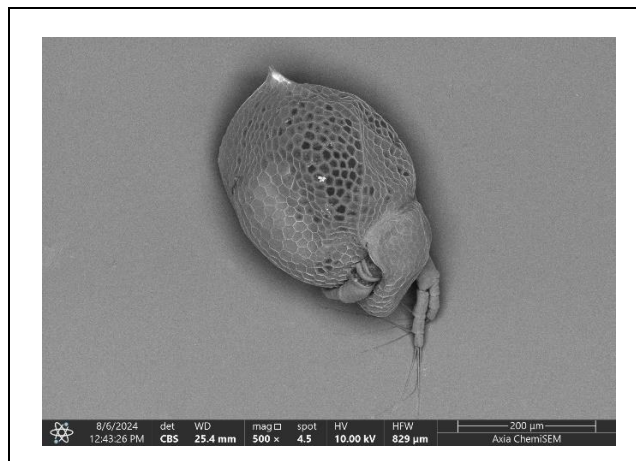


Image 1: Scanning Electron Microscopy Image of *Daphnia magna*.

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Dilution Water used for study

Deionised water with an enrichment of certain minerals is used in the test.

Dilution water composition

Parameter	Concentration in mg/L
CaCl ₂ *2H ₂ O	293.80
MgSO ₄ *7H ₂ O	123.30
NaHCO ₂	64.80
KCl	5.80
Resulting hardness in mmol/L	2.50
Resulting hardness in mg CaCO ₃ /L	250

After preparation, the dilution water is aerated, and the pH is adjusted to 7.8 + 8.2.

Instruments and Devices

The following instruments and devices were used in the performance of the study -

Thermometer for temperature, Adjustable pipettes, Glass measuring cylinders, Glass measuring flasks, pH-meter
DO meter, COD digester, CHNS analyser.

Test protocol

The newborn Daphnia, aged between 0 and 23 hours was separated from 24 hrs old culture and immediately placed into a 100 mL-beaker containing dilution water.

Animals which showed no apparent damage were used for the test.

Using a glass tube the Daphnia were caught and lifted from the beaker.

The four groups of each test concentration were taken and 5 daphnids are inoculated per group resulting in 20 Daphnia per test concentration and control.

The test vessels were left to stand for 48 hours.

After 24 and 48 hours, the immobilised Daphnia were counted. Daphnia are considered immobilised if they do not perform any movements or are only able to move their antennae when the beaker is gently agitated.

The pH and the concentration of dissolved oxygen in the test vessels were measured at the beginning and at the end of the test.

Experimental Conditions

Test vessels	Glass beakers, nominal volume 100 mL
Feedings	None
Temperature	20 ± 2 °C
Duration	48 hours
Observation times	24 and 48 hours
Replicates	four vessels, each containing test solution and five daphnia
Control	four vessels, each containing dilution water and five daphnia

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Observation:

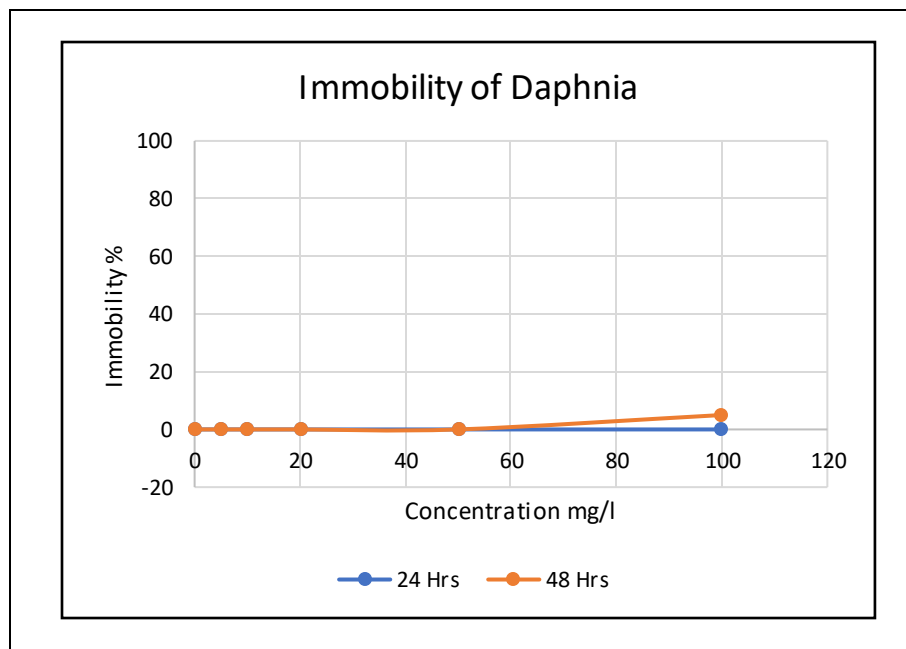
A. Immobilities of *Daphnia magna*

In the control and test concentration none of the daphnia died or showed any signs of abnormal behaviour throughout the test period. The 50% immobilisation was observed at 2.0mg/L concentration in Reference control.

Table 1: Immobilities of *Daphnia magna*

Nominal Concentration in mg/L	Total number of Daphnia exposed	Immobility 24 Hours					Immobility 48 hours				
		Observation per group			% Immobilisation	Observation per group			% Immobilisation		
Blank Control (with emulsifier polysorbate 80)											
Control	20	0	0	0	0	0	0	0	0	0	0
Reference Control											
0.5	20	0	0	0	0	0	NA	NA	NA	NA	NA
1.0	20	1	0	0	0	5	NA	NA	NA	NA	NA
2.0	20	3	1	4	3	55	NA	NA	NA	NA	NA
3.0	20	4	4	3	3	70	NA	NA	NA	NA	NA
5.0	20	5	5	4	3	85	NA	NA	NA	NA	NA
Test Sample											
5	20	0	0	0	0	0	0	0	0	0	0
10	20	0	0	0	0	0	0	0	0	0	0
20	20	0	0	0	0	0	0	0	0	0	0
50	20	0	0	0	0	0	0	0	0	0	0
100	20	0	0	0	0	0	1	0	0	0	5

GRAPH: 1: Immobility of *Daphnia magna* after 24 and 48 hours of exposure to test concentrations.

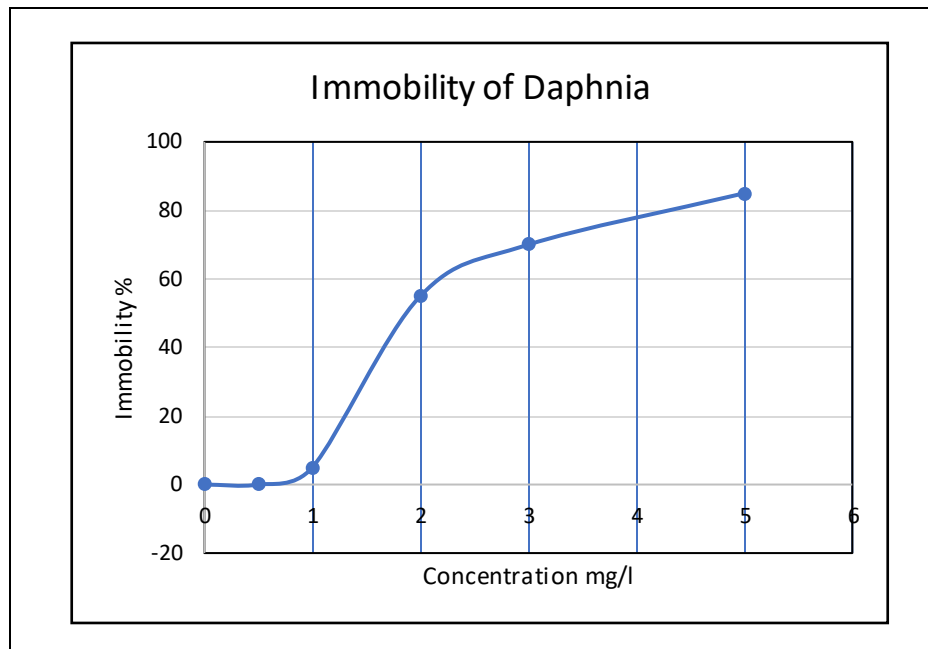


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GRAPH 2: Immobility of *Daphnia magna* after 24 hours of exposure to reference material.



B. Confirmatory Chemical Analysis

Compound of Test substance was analyzed at initial and after 48 hrs of test duration using Gas Chromatography Mass – Spectrometry Detection (GC – MS)

Table 2: Initial test compound determination by GC – MS.

Sr No	Initial Test Sample
1	Demecolcine
2	2-(Heptanoyloxy)propane-1,3-diyl dioctanoate
3	1H,5H,9H,11H,15H-xantheno[2,3,4-ij:5,6,7- i'j']diquinolizine, 2,3,6,7,12,13,16,17-octahydro-9-(trifluoromethyl)-
4	1,5-Dioxo-8,11-diazacyclotridecane-7,12-dione, 8,11-dihexyl-3,3-dimethyl-
5	4,11-Dihexyl-16,16-dimethyl-1,14-dioxo-4,11- diazacycloheptadecane-3,12-dione
6	(N,N'-Bis(3- butoxysalicylidene)ethylenediaminato)cobalt(ii)
7	l-Leucine, N-methyl-n-pentadecafluorocarbonyl-, octyl ester
8	2-[.gamma.-Dimethylaminopropylamino]-3- methyl-4-diethylamino-5-[4-chlorophenyl]-8- chloro-9-methoxybenzo[h][1,6]naphthyzidine
9	Colchicine, N-desacetyl-N-[(2',2',5',5'-tetramethyl- N'-oxy-pyrrolid-3'-yl)carbonyl]-
10	4,11-Dihexyl-16,16-dimethyl-1,14-dioxo-4,11- diazacycloheptadecane-3,12-dione
11	(N,N'-Bis(3- butoxysalicylidene)ethylenediaminato)cobalt(ii)

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Graph 3: Initial test compound determination by GC – MS.

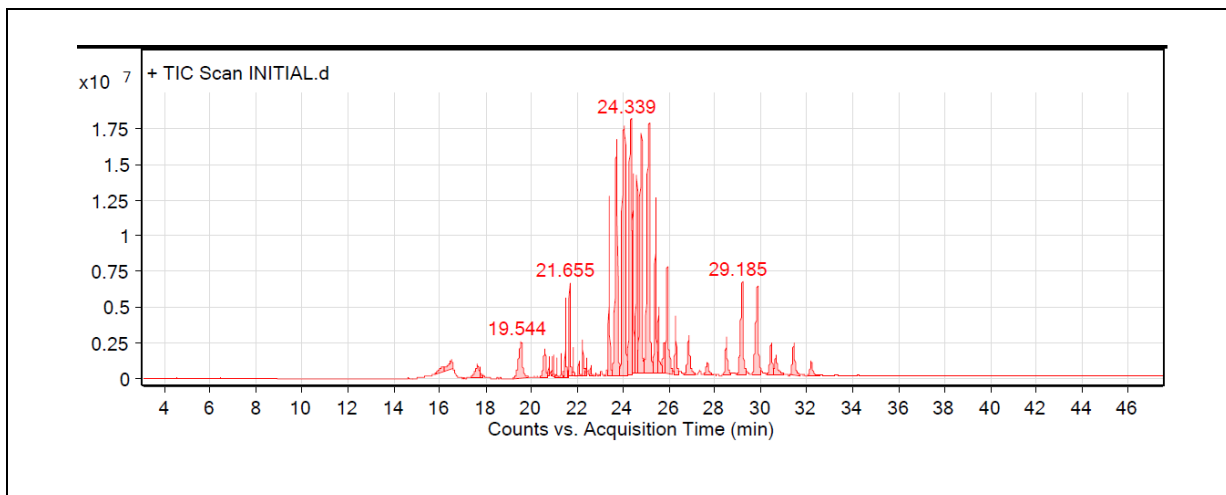


Table 3: Test compound after 48 hours determination by GC – MS.

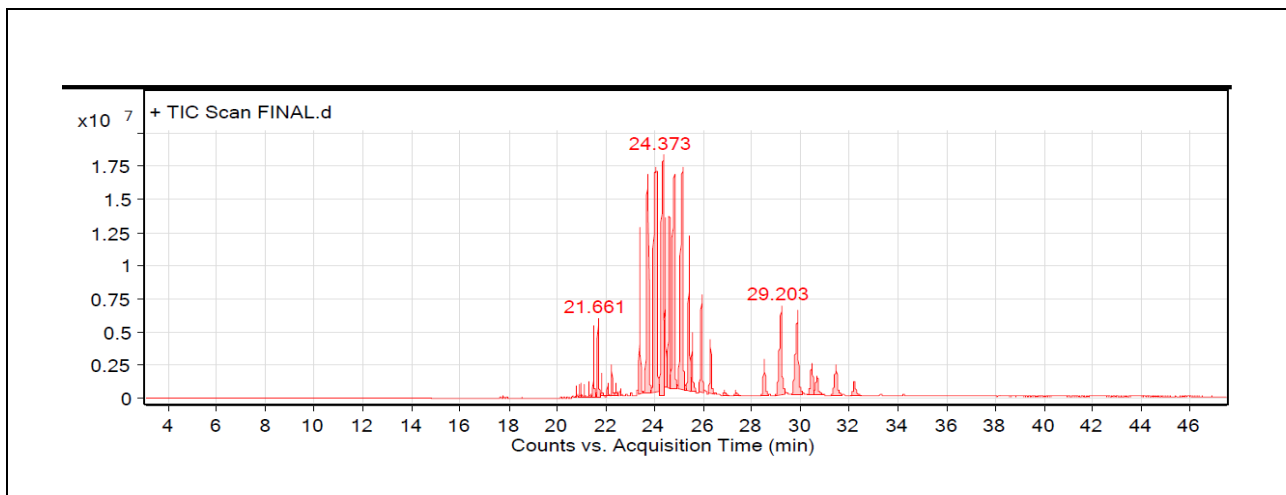
Sr No	Test Sample after 48 Hrs
1	Demecolcine
2	2-(Heptanoyloxy) propane-1,3-diyl dioctanoate
3	1H,5H,9H,11H,15H-xantheno[2,3,4-ij:5,6,7- i'j']diquinolizine, 2,3,6,7,12,13,16,17-octahydro-9- (C26H27F3N2O)-
4	1,5-Dioxa-8,11-diazacyclotridecane-7,12-dione, 8,11-dihexyl-3,3-dimethyl-
5	4,11-Dihexyl-16,16-dimethyl-1,14-dioxa-4,11 diazacycloheptadecane-3,12-dione
6	(N,N'-Bis(3- butoxysalicylidene)ethylenediaminato)cobalt(ii)
7	l-Leucine, N-methyl-n-pentadecafluorocarbonyl-, octyl ester
8	2-[.gamma.-Dimethylaminopropylamino]-3- methyl-4-diethylamino-5-[4-chlorophenyl]-8- chloro-9-methoxybenzo[h][1,6]naphthyzidine
9	Colchicine, N-desacetyl-N-[(2',2',5',5'-tetramethyl- N'-oxy-pyrrolid-3'-yl)carbonyl]-
10	4,11-Dihexyl-16,16-dimethyl-1,14-dioxa-4,11- diazacycloheptadecane-3,12-dione
11	(N,N'-Bis(3- butoxysalicylidene)ethylenediaminato)cobalt(ii)

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Graph 4: Test compound after 48 hours determination by GC – MS.



C. pH and Dissolved oxygen observations during analysis

Table 4: pH and O₂ observation for test sample and reference control

Nominal Concentration in mg/L	pH		O ₂ – Concentration in mg/L	
	0h	48 h	0 h	48 h
Blank control (with emulsifier polysorbate 80)				
Control	7.78	7.86	8.0	7.8
Reference control				
0.5	7.10	7.23	8.0	7.9
1.0	7.14	7.25	8.0	7.9
2.0	7.08	7.11	8.0	7.9
3.0	7.05	7.56	8.0	7.9
5.0	7.05	7.34	8.0	7.9
Test Sample				
5	7.75	7.65	7.9	7.7
10	7.78	7.73	7.9	7.7
20	7.98	7.77	7.9	7.7
50	7.98	7.80	7.9	7.7
100	7.96	7.81	7.9	7.7

D. Biological Results Test Sample and Reference Control

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Table 5: Estimation of EC 50 of Test Sample and Reference Control

Parameter	Value	95% - Confidence interval
Reference Control		
24h NOEC	0.5 mg/L	n. d.
24h EC 50	2.0 mg/L	n. d.
Test Sample		
24h NOEC	100 mg/L	n. d.
48h NOEC	50 mg/L	n. d.
24h EC 50	>100 mg/L	n. d.
48h EC 50	>100 mg/L	n. d.

n. d. – Not determinable

Conclusion

The 24-Hour and 48-Hour acute toxicity of TRANSOL SYNTH.100 to the freshwater planktonic crustacean, *Daphnia magna*, was determined in accordance with the requirements of OECD Chemicals Testing Guideline No. 202 *Daphnia* sp. Acute Immobilisation Test. This test was conducted using the following nominal loading rates: 5.0, 10, 20, 50 and 100 mg test item/L in addition to a control.

A range-finding test was performed until 1000 mg/L to determine the range of concentrations for the definitive test, as no Immobilisation was observed limit test was performed using 5 mg/L to 100 mg/L. No effect on *Daphnia magna* was observed.

The following results could be determined for the test item TRANSOL SYNTH.100 (species: *Daphnia magna*)

24 h-NOEC = 100 mg/L

48 h-NOEC = 50 mg/L

24 h-EC50 = >100 mg/L

48 h-EC50 = >100 mg/L

The validity criteria for control immobility ($\leq 10\%$) and dissolved oxygen (≥ 3 mg/L) were both satisfied. The test is therefore considered valid.

Authorized Signatory



Akshata Rangle

Assistant Manager- Microbiology

End of Report

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